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[22CSE103]

SECOND SEMESTER B.Tech EXAMINATION, APRIL - 2023

COMPUTER NETWORKS

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ii. **Answer All Questions**

PART-A **05 X02=10**

1.	**a.**	Discuss the services provided by DNS.	L1	CO1	**02**
	b.	What is RED Algorithm? List out the components of RED algorithm.	L1	CO2	**02**
	c.	Compare Static and Dynamic IP Addressing.	L2	CO3	**02**
	d.	Write the applications of DHCP.	L2	CO5	**02**
	e.	List out the issues in Wi-Fi.	L2	CO6	**02**
		PART-B		**06 X 05=30**	
2.	**a.**	Define network. With a neat diagram, explain the basic topologies.	L2	CO2	**05**
		OR			
	b.	Explain HTTP request and response message formats with diagrams.	L2	CO2	**05**
3.	**a.**	With a neat diagram, explain TCP state diagram.	L2	CO3	**05**
		OR			
	b.	Discuss the purpose of the UDP header. Explain UDP Header format with a neat diagram.	L2	CO3	**05**
4.	**a.**	Describe different types of network devices commonly found within a network.	L2	CO4	**05**
		OR			
	b.	Differentiate between IPv4 and IPv6 addressing.	L2	CO4	**05**
5.	**a.**	How does CSMA/CD handle collisions in a network? Explain.	L2	CO5	**05**
		OR			
	b.	Explain the common challenges in configuring and maintaining VLANs in a network?	L2	CO5	**05**
6.	**a.**	Illustrate pure aloha and slotted aloha with neat diagram.	L2	CO5	**05**

		OR			
	b.	With a neat diagram, explain 802.11 Mac frame format	L2	CO6	**05**
7.	**a.**	What is Bluetooth? Explain Piconet and Scatternet in detail.	L2	CO6	**05**
		OR			
	b.	Describe the different types of Wireless Topologies.	L2	CO6	**05**
		PART-C	**06 X 10=60**		
8.	**a.**	What is Data communication? With a neat diagram, explain the components of data communication and also, brief out the different ways of representing data.	L2	CO1	**10**
		OR			
	b.	What is Addressing? Explain different types of addresses in TCP/IP. Describe the role of protocol layering with its advantages and disadvantages.	L2	CO1	**10**
9.	**a.**	Explain the DNS working principle, resource records and message format.	L2	CO2	**10**
		OR			
	b.	Explain Karn's Algorithm in detail. Consider the following contents of a UDP header in hexadecimal format and answer for the following: **CB84000D001C001C** i. What is the source port number? ii. What is the destination port number? iii. What is the total length of the user datagram? iv. What is the length of the data? v. Is the packet directed from a client to a server or vice versa?	L2	CO2	**10**
10.	**a.**	Explain the following in detail: i. SMTP ii. FTP	L2	CO3	**10**
		OR			
	b.	With neat block diagram, describe the OSI reference model with functionalities of each Layer.	L2	CO3	**10**

11.	**a.**	What is Distance Vector Routing? Write an algorithm for the same and explain with an example.	L2	CO4	**10**
		OR			
	b.	Define Enhanced Interior Gateway Routing Protocol. Explain the messages and tables to communicate with the neighbor devices that operate EIGRP.	L2	CO4	**10**
12.	**a.**	What is CSMA? Illustrate Space model of Collision, Vulnerable Time and persistent methods.	L2	CO5	**10**
		OR			
	b.	What is VLAN? Explain its features, memberships, VTP in detail.	L2	CO5	**10**
13.	**a.**	What are IEEE 802.11 networks? Explain its components and frame format.	L2	CO6	**10**
		OR			
	b.	Write a brief note on: i. NAT ii. Splitting Algorithms	L2	CO6	**10**